

Topic: Joint probability Distribution

Example:

- A) Joint probability function
- B) Joint probability table
- C) $P(X+Y \leq 1)$
- D) Marginal distributions of X and Y
- E) Mean of X and Y
- F) Var (X) and Var (Y)
- G) Cov(X,Y)
- H) Correlation (X,Y)
- I) Statistical Independence
- J) Conditional probability of $X/Y=2$

Solution:

A) Joint Probability Function:

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}},$$

for $x = 0, 1, 2; y = 0, 1, 2$; and $0 \leq x + y \leq 2$.

B) Joint Probability Table:

$f(x, y)$		x			$h(y)$
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$g(x)$		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

C)

The probability that (X, Y) fall in the region A is

$$\begin{aligned} P[(X, Y) \in A] &= P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) \\ &= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14}. \end{aligned}$$

D)

Marginal distribution of Y

Y	$h(y)$	$Y h(y)$	$Y^2 h(y)$
0	$15/28$	0	0
1	$3/7$	$3/7$	$3/7$
2	$1/28$	$2/28$	$4/28$

Marginal distribution of X

X	$g(x)$	$X g(x)$	$X^2 g(x)$
0	$5/14$	0	0
1	$15/28$	$15/28$	$15/28$
2	$3/28$	$6/28$	$12/28$

E) $U_x = E(X) = \sum x g(x) = 0 + 15/28 + 6/28 = 21/28$

$U_y = E(Y) = \sum y h(y) = 14/28$

F) $\text{Var}(X) = E(X^2) - (E(X))^2 = 27/28 - (21/28)^2 = 45/112$

Where $E(X^2) = \sum X^2 g(x) = 27/28$

$\text{Var}(Y) = E(Y^2) - (E(Y))^2 = 4/7 - (14/28)^2 = 9/28$

G) $\text{Cov}(X,Y) = E(X,Y) - E(X) E(Y) = 3/14 - 3/4 * 1/2 = -9/56$
 $E(X,Y) = \sum \sum XY f(x,y) = 0 + 0 + 0 + 0 + 3/14 + 0 + 0 + 0 + 0 = 3/14$

		x			$h(y)$
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
$g(x)$		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

If $E(x,y) = E(x) E(y)$ (Independent)

$\sigma_{xy} = \text{Cov}(x,y) = 0$ (no correlation) (Independent) Range of correlation (-1 to 1)

H) Correlation = $\text{cov}(xy) / (sd(x) * sd(y))$ where $\sigma x = sd(x), \sigma y = sd(y)$
 $= -9/56 / (0.6339 * 0.5669) = -1/\sqrt{5} = -0.44$ (Negative moderate correlation)

I) Independence

$E(x,y) = E(x) E(y)$ (Independence)

$f(x,y) = g(x) h(y)$ (independence) (this equation must hold for every pair of x,y)

$f(0,0) = g(0) h(0)$

$3/28 = (5/14) (15/28)$

$0.1071 \neq 0.1913$ (X and Y are statistically not independent)

Example 3.21: Show that the random variables of Example 3.14 are not statistically independent.

Proof: Let us consider the point (0,1). From Table 3.1 we find the three probabilities $f(0,1)$, $g(0)$, and $h(1)$ to be

$$f(0,1) = \frac{3}{14},$$

$$g(0) = \sum_{y=0}^2 f(0,y) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$h(1) = \sum_{x=0}^2 f(x,1) = \frac{3}{14} + \frac{3}{14} + 0 = \frac{3}{7}.$$

Clearly,

$$f(0,1) \neq g(0)h(1),$$

and therefore X and Y are not statistically independent. └

(J) Conditional Probability of X given $Y=2$, $P(X/Y=2)$

Definition 3.11: Let X and Y be two random variables, discrete or continuous. The **conditional distribution** of the random variable Y given that $X = x$ is

$$f(y|x) = \frac{f(x,y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of X given that $Y = y$ is

$$f(x|y) = \frac{f(x,y)}{h(y)}, \text{ provided } h(y) > 0.$$

Now, when X is random and Y is fixed:

$$P(X/Y=2) = f(x,y)/p(y=2)$$

$$= f(0,2)/p(y=2) = 1/28 / 1/28 = 1$$

$$= f(1,2) / p(y=2) = 0$$

$$= f(2,2)/p(y=2) = 0$$

If Y is fixed and X is random:

$$P(Y=1/X) = f(x,y) / g(x)$$

$$= f(0,1)/g(0) = (3/15) / (5/14) = 0.56$$

$$= f(1,1)/g(1) = (3/14) / (15/28) = 0.4$$

$$= f(2,1) / g(2) = (0) / (3/28) = 0$$

If X and Y both are fixed, then:

$$P(Y=1/X=1) = f(1,1)/g(1) = 3/14 / 15/28 = 0.4$$